ACIT 4850 – Enterprise Systems Integration – Assignment 3

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Total Marks	18
Due Dates	End of last class in Week 15: • April 14th for Sets B and C • April 15th for Set A

This assignment is to be completed and submitted with your partner (i.e., Group submission) if applicable.

Overview

For this assignment you will be deploying your Point and CarLot applications to a separate test environment and leveraging shared libraries in Jenkins

1. Setup Pipelines

In the ci_functions project in your GitLab installation, create a copy of your python_build.groovy shared library. You can call it assignment3.groovy.

In your point and carlot projects in GitLab, create new Jenkinsfile_assignment3 pipelines in the ci folder that use the shared library defined in assignment3.groovy.

In Jenkins, create new pipeline jobs for each of the two new Jenkinsfiles.

You will be modifying the assignment3.groovy shared library with a couple of new stages shortly.

2. Setup VM

Create a new small VM in Azure with Ubuntu as the OS, ideally in the free tier. Install docker and git on the VM.

In your GitLab installation, create a new project called deployment with a docker-compose.yml file. Configure the Docker Compose files with your two services: point and carlot. They should expose the correct ports for each service (as per Lab 11) and should refer to the images on DockerHub (not local images or dockerfiles).

Clone this new project on your VM. You should be able to bring up your two services using docker compose up -d and bring them down with docker compose down.

3. Pipeline Updates

Add two new stages to your pipeline (in assignment3.groovy) in the most appropriate location: **Security** and **Deploy**

Security

Create a new stage called Security which either:

- Scans your docker image for potential vulnerabilities (**if your Group # is even**)
- Scans your Python code for potential vulnerabilities (**if your Group # is odd**)

Make sure to put the stage at the most appropriate place in your pipeline.

Deployment

Create a new stage called deployment (or replace that stage if you've already done it Lab 11) that:

- SSHs into your VM that your created in part 4 above. You will need to setup an SSH credential on the VM and in Jenkins to do this.
- Pull the latest images from DockerHub (using a docker compose command)
- Start/restart your services (using a docker compose command). If they aren't running it should start them. If they are running, it should only restart a service if its corresponding image has been updated.

This stage should only run if a DEPLOY pipeline parameter is set (as per Lab 11).

Note: For extra practice (and full marks in this assignment), update the fail-under threshold in the Lint stage such that it fails either the new point or carlot pipelines and address those issues so it passes again. Make sure you remember what code issues you fixed for your demo.

4. Testing and Demonstration

Before demonstrating this to your instructor, verify the following:

- A change you make to the code is reflected in the deployment if the pipeline is run and the DEPLOY is enabled. You will not get full marks in your demo if it doesn't.
- You can make a change to your code (or dockerfile dependencies) to get your Security stage to fail.

Demonstration:

- Make sure your services are down on the VM.
- Run each of your pipelines successfully without DEPLOY enabled.
- Run one of your pipelines unsuccessfully with a security issue that causes the Security stage to fail.
- Run one of your pipelines successfully with DEPLOY enabled. Your services on your VM should be automatically started. Also:

    - Confirm in the Jenkins Console Output that you are using a credential for SSH (and it is hidden)
    - Confirm in the Jenkins Console Output that your services have started
- Make a harmless code change that has some impact on the output from one of your services. Run the corresponding pipeline successfully with DEPLOY enabled. You should be able to show that the change is in the deployment.

Submission

- (10 marks max) Demonstration – For the five items above.
- (6 marks) Submit the following:
    - (2 marks) assignment3.groovy
    - (2 marks) Jenkinsfile_assignment3 (for both point and carlot)
    - (2 marks) docker-compose.yml file
    - (1 mark) Console output for the Deployment stage
- (1 mark) For correcting issues reported by pylint by fixing them in the Python code.